

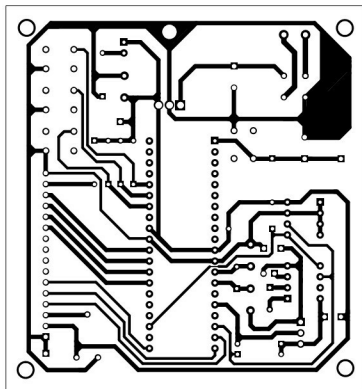


Fig. 3: An actual-size, single-side PCB for real-time clock with temperature logger

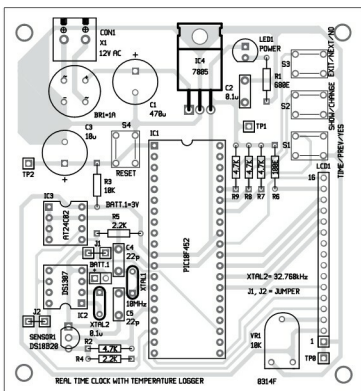


Fig. 4: Component layout for the PCB

language and compiled using MPLAB IDE. MPLAB IDE is a compiler for PIC. The program is burnt in the microcontroller using a suitable PIC programmer. Following fuse bytes are forced through source code:

```
#pragma config OSC=HS, OSCS=OFF
#pragma config PWRTE=ON, BOR=ON, BORV=45
#pragma config WDT=OFF, LVP=OFF
#pragma config DEBUG=OFF, STVR=OFF
```

Construction and testing

An actual-size, single-side PCB for the real-time clock with temperature logger circuit is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on the PCB to minimise assembly errors. Use IC base for microcontroller IC. To test the circuit for proper functioning, verify the voltage levels as shown in the test points table.

When the system is turned on for the first time, it shows a welcome message, and after that it shows screen like the one in Fig. 5 because

RTC is off. We now have to set time in it. To set the time, press any of the switches and the display will show options as in Fig. 6.

Now press switch S1 and the display changes to as shown in Fig. 7.

You can navigate the cursor by

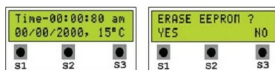


Fig. 5: Display when RTC is off

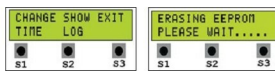


Fig. 6: Time setting menu



Fig. 7: After pressing S1



Fig. 8: Write in RTC



Fig. 9: EEPROM erasing menu



Fig. 10: Erasing EEPROM in progress

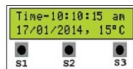


Fig. 11: Time, date and temperature display



Fig. 12: Logged data display

pressing switch S1 for left and S3 for right. To change a digit, press switch S2. After setting all the digits, press S3 again to write this time in RTC as shown in Fig. 8.

Now press S1 or S3 as your wish. If you choose S1, EEPROM will be cleared. Refer Fig. 9.

It will take a couple of seconds to clear the EEPROM. After clearing the EEPROM, the system will display time, date and temperature as shown in Fig. 11.

To see logged temperature, press any key while the system is showing time and date. It will show the display as in Fig. 6. Now choose S2 for SHOW LOG option; it will show most recent logged data as in Fig. 12.

This means:

Date of log: 17/01/14

Minimum temperature: 14°C

Maximum temperature: 15°C

You can see other logged data of previous or next day by pressing S1 or S2 switches. Press S3 switch to exit from log. ●

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EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on source.efymag.com website.